Vacuum Evaporator



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Vacuum evaporator

What it is used for

A vacuum evaporator removes solvent or water from a liquid by evaporation under reduced pressure. It is used to concentrate compatible products at lower temperatures while controlling heat and vapour removal.

Main parts

- Evaporation vessel or boiling flask
- Vacuum pump and vacuum line
- Condenser and vapour collection receiver
- Heating bath or jacket
- Temperature and vacuum pressure gauges
- Seals, valves, and control panel

Before you start

1. Read the operating instructions and verify the liquid is compatible with the equipment and seals.
2. Inspect the vessel, glassware if fitted, hoses, clamps, seals, and condenser for cracks or leaks.
3. Confirm the vacuum pump, cooling supply, and condensate receiver are ready for operation.
4. Check that the vessel is rated for vacuum and that all connections are secure.
5. Confirm the liquid volume is within the vessel's working capacity and is unlikely to foam excessively.
6. Wear suitable eye protection, gloves, and any additional protection required by the liquid's safety data sheet.

How to use

1. Load the approved liquid into the evaporation vessel without exceeding the recommended fill level.
2. Secure the vessel and confirm that the condenser and collection receiver are connected correctly.
3. Start cooling water or the approved condenser cooling system before heating.
4. Switch on the vacuum pump and reduce pressure gradually according to the procedure.
5. Set the heating temperature and vacuum level appropriate for the product and solvent.
6. Monitor temperature, pressure, evaporation rate, foaming, and condensate collection throughout the run.
7. Adjust heating or vacuum gradually if bumping, foaming, or unstable boiling occurs.
8. When the target concentration is reached, stop heating and allow the system to cool as required.

9. Release vacuum slowly through the designated valve before opening the vessel or disconnecting components.
10. Switch off the pump and cooling system in the prescribed sequence, then clean and inspect the equipment.

⚠ Safety

- Never use cracked glassware or a vessel that is not rated for vacuum.
- Use a protective shield when required for glass vessels operating under vacuum.
- Do not open the system while it is under vacuum or while hot solvent remains inside.
- Use suitable ventilation and keep flammable solvent vapours away from ignition sources.
- Verify solvent compatibility with the pump, seals, tubing, and condenser.
- Isolate electrical power before maintenance and follow procedures for hazardous solvents.

Care & storage

- Clean the vessel, condenser, and receiver using methods compatible with the processed liquid.
- Inspect seals, vacuum tubing, clamps, and glassware regularly for wear or damage.
- Maintain the vacuum pump and change its oil or filters according to the manufacturer's instructions.
- Record operating conditions, cleaning, and maintenance where required.

Common mistakes to avoid

- Applying vacuum too quickly can cause bumping, foaming, or liquid carryover.
- Heating without adequate condenser cooling can release solvent vapours.
- Overfilling the vessel can contaminate the vacuum line and collection system.
- Releasing vacuum suddenly or opening hot equipment can cause splashing and injury.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.